

Ana Lorena Ortiz Loyola

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Data science student (Tecnológico de Monterrey, expected 2027) with hands-on experience in data engineering, machine learning, and cryptography research. Built automated data quality and migration testing at Deutsche Bundesbank (AnaCredit), and independent projects spanning predictive modeling, classification, and lattice-based cryptography. Comfortable owning a data pipeline end-to-end, from ingestion through auditable documentation.

Core Competencies

- **Programming:** Python (Pandas, NumPy, Scikit-Learn, Keras, TensorFlow), R, Java, C++
- **Data & Databases:** SQL, SQLite, data cleaning and pipeline design, Excel
- **Machine Learning:** Random Forest, SVM, K-Means, predictive modeling, classification, clustering
- **Cryptography & Security:** Lattice-based schemes (LWE, NTRU), complexity and security parameter analysis
- **Cloud & Tools:** Azure Cognitive Services, AWS, Jira, Git, Tableau, Matplotlib, Seaborn

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to ensure identical business logic after data migration
 - Implemented automated error detection identifying deviations (duplicate IDs, value-range violations)
 - Automated Jira ticket creation and emails to reporting banks for detected data quality issues
 - Significantly reduced response time to data quality problems
 - Worked within an agile Scrum framework (sprint planning, daily stand-ups, sprint reviews)

Automated Validation Framework for Regulatory Reporting Data – converted a non-scalable, case-by-case validation script into a parameterized, reusable Python solution

 - Designed parameterized MySQL queries applicable to any database, attribute, and reporting period, replacing hard-coded per-case script logic
 - Shifted computationally intensive operations to the database layer (SQL views, aggregated queries) to efficiently process large regulatory databases
 - Built a YAML-driven configuration interface enabling batch validation across multiple databases, attributes, and reporting periods in a single call
 - Extended the validation logic with percentile-based statistical benchmarking over a rolling 12-month window to flag anomalous reporting frequencies and separate false positives from genuine data-quality issues
 - Added comprehensive exception handling for connection failures, invalid input, and edge cases such as composite primary keys

Test Data Generator for the Bundesbank Platform – built a configuration-driven system producing complete, consistent synthetic test datasets

 - Generated Cartesian-product attribute combinations per table from a custom template, independent of data type (strings, booleans, numbers, dates)
 - Maintained consistent IDs across tables to preserve referential integrity in generated data
 - Exported datasets as XML with full schema metadata, referential-integrity information, and plausibility metrics (e.g. expected record counts)

Projects

- **Tecnológico de Monterrey** 10/2024–02/2025
Lattice-Based Cryptography Research
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python
 - Contributed to a scientific publication analyzing security parameters and runtime behavior across lattice dimensions
- **Tecnológico de Monterrey** 10/2024–12/2024
Predictive Modeling – Healthcare Optimization
 - Analyzed 25+ years of Mexican national health data to identify patient risk profiles and resource bottlenecks
 - Built predictive models (Random Forest, Neural Networks) and K-Means clustering for high-risk segmentation
- **Personal Project (GitHub)** 05/2026
Banking Data Pipeline – Data Cleaning & Audit Log
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts) with a normalized SQLite database and foreign-key integrity checks
 - Staged data cleaning with a fully auditable log; automated balance maintenance via SQL triggers
- **Tecnológico de Monterrey** 04/2023–06/2023
Mexico City Air Quality Analysis – SVM Classification
 - Analyzed city-wide NO_x/O₃ data and applied SVM models (linear & RBF) to classify critical ozone levels with high accuracy
 - Correlation analysis and visualizations identified elevated pollution in southern city districts; findings presented in a final presentation

Education

- **Tecnológico de Monterrey** **Mexico**
2022–2027
BSc in Data Science
 Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: data analysis & statistics, machine learning & AI, cryptography & security, NLP, numerical optimization.
- **Universität Göttingen** **Germany**
2025–2026
Exchange Semester
- **Tecnológico de Monterrey** **Mexico**
2019–2022
Abitur
 Grade: 1.0. National Physics Olympiad 2020; Honorable Mention, Physics Olympiad 2021.

Languages

- Spanish (native), English (C1, TOEFL iBT 105/120), German (C1).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.